BCG X

Data Science

Understand the business and frame the problem

Getting to know the client and their challenges

Know what you need

While you were reviewing the case materials, your team held a team meeting to discuss the client brief and generate hypotheses about the cause.

During the meeting the someone on your team noted that the client strongly believes that the for churn (among others) is that customers have become more "sensitive" to the price and are opting toward low-cost providers – we need to test this hypothesis.

You will partner with Estelle, a senior data scientist, to uncover the insights we need to validate or disprove the idea that price is the critical factor.

You and Estelle know that the first step in any case is to **understand the business and frame the problem** – we need the right data points and techniques to do that.

To ensure we can successfully identify the cause, your task is to draft an email clearly outlining the ***data that we'll need from the client*** and the ***techniques*** you will use to ***investigate the issue***.

To help get you started, ask yourself:

1. What are all of the possible reasons a customer would choose to stay with or switch energy providers? (for example: price, energy source, customer service, etc.)
2. What data points would be needed in order to effectively analyze causes of churn? (e.g. purchasing trends, location of business etc.)

3. Once we have the data, what steps will you take to process and analyze the information? (hint: use the 5-step process outlined previously)

Use the text field below to write your email – good luck!

Your response

Hi Estelle,

I have outlined my thinking on the data we will need from the client and the techniques I would propose to test the price sensitivity hypothesis.

Price is only one possible driver of churn. We should also account for other factors like contract terms, tenure, consumption patterns, customer margin, and competitor activity so we can isolate price's actual effect.

Data to request from PowerCo:

- Customer Information: tenure, contract type and dates, subscribed power, firm characteristics (industry, size, location etc.).
- Consumption Data: historical and forecasted electricity and gas usage per customer, ideally broken down by period.
- Pricing data: historical prices per customer by period(peak, off-peak, mid-peak)
- Financial Data: net margin and any forecasted revenue per customer
- External Factors: like competitor pricing, market rates, to see how PowerCo's price compares.

Proposed Approach:

- Define the churn precisely and translate the price-sensitivity belief into a testable hypothesis.
- Gather the datasets above and confirm coverage, and time range with the client.
- Clean and merge the data, handle missing values and outliers, and examine distributions and correlations

- Construct price-sensitivity metrics (like y-o-y) alongside consumption, tenure, and margin features.
- Build a predictive churn model and assess the importance of price features relative to other drivers, so that we can explain how much price actually explains.

This should let us either validate or disprove the price hypothesis with evidence rather than assumption. Happy to discuss and refine before we take it to the client.

Best,
Anuj Khadka

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